

Architecture — Network Attack Forecasting (SIH26153)

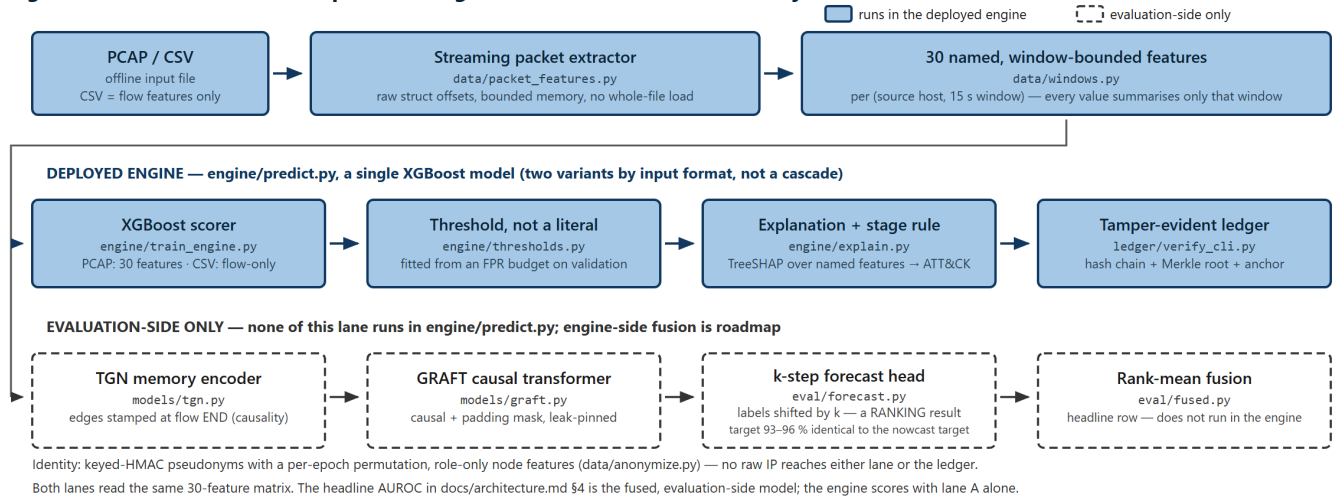
Two pages. Numbers are reproducible via `python scripts/make_ablation_table.py`.

1. Problem and unit of prediction

Given network traffic up to time t , forecast the **infiltration probability and ATT&CK stage at $t+k$** , explain each forecast in named features, and commit it to a tamper-evident ledger — fully offline. The unit is **(source host, 15 s window)** on a 5 s stride, never per-flow and never whole-network; the network-level score is the **max** over host scores in a window. Success is measured by **lead time** (seconds between the first alert on the attacking host and the annotated attack completion) at a fixed false-positive budget — not by F1.

2. Pipeline

Figure 1 — Data flow: what ships in the engine, and what is evaluation-only



docs/img/02-model-stack.svg carries the same split per component with each one's measured status, including the RSSM world model — a negative result that does not ship.

Ingest & features. A single streaming extractor (data/packet_features.py) parses pcap bytes with raw struct offsets (~110k packets/s, bounded memory on a 4.3 GB flood capture) and emits both flow records and per- (src, window) packet statistics: TTL mean/variance, TCP window, fragment flags, an 8-bin payload histogram, distinct destination ports/IPs, port entropy, a **sequential-vs-randomised scan ratio**, and retransmission counts (tshark-verified, scapy fallback). data/windows.py joins these into a **30-feature, strictly window-bounded** matrix — every value summarises only packets whose own timestamp lies in the window. That constraint is load-bearing: attributing whole-flow totals to a flow's start window leaked forecast-horizon traffic into the present and would have fabricated the headline metric (docs/decisions/003).

Identity. Nodes are keyed-HMAC pseudonyms with a per-epoch permutation; node features are role-only (internal/external, hashed /24). No raw IP reaches the model or the ledger.

Encoder. models/tgn.py maintains a TGN memory over the host graph; edges are flow events timestamped at **flow end** (a flow's statistics are only known once it completes — the same causality rule as the features). Each host's memory vector at a window boundary is its embedding. Memory is reset between splits so training state never reaches test.

Sequence model. models/graft.py is a 2-layer causal Transformer over per-host window sequences — always masked with both a causal mask and a padding mask, verified *bit-identically* by test_no_future_leakage (perturbing windows $t+1...T$ leaves the prediction at t unchanged). It carries a Dirichlet evidential head (uncertainty = K/S) and a benign-only reconstruction head as an OOD signal.

Forecast head. Labels are shifted by k , so the model predicts "attack on this host in k windows". This is the shipped M7 deliverable; the planned RSSM rollout failed its gate and is recorded as a negative result (docs/decisions/004, docs/limitations.md §4).

3. Engine, explainability, ledger

engine/predict.py runs fully offline from persisted artefacts: it builds features from the input file, scores each host-window against a threshold **fitted from an FPR budget on validation** (never a literal), and for each alert emits a JSON-schema-validated object. The deployed scorer is a **single XGBoost model** (CPU-cheap and natively TreeSHAP-explainable) — one of two variants selected by input format, not a cascade: the full 30-feature model for PCAP, a flow-only model for CSV. The fused TGN+XGB headline in §4 is the **eval-side** model and does not run in the engine; engine-side fusion is roadmap. Each alert object carries: probability, stage, MITRE technique, top-5 **named** features, the top-3 contributing windows (where the attack was forming), and the flagged flows in that window. Attributions are TreeSHAP over the deployed model in named-feature space — the problem statement rules out black-box output, so explanations name payload_hist_0 or distinct_dst_ips, never an embedding index. engine/technique_map.yaml maps stage + observed pattern to techniques actually evidenced in our data (internal scan → **T1046**, credential guessing → **T1110**, C2 → **T1071**, availability attack → **T1498**, bulk egress → **T1048**).

Every forecast is appended to an **append-only hash-chained JSONL** with per-batch Merkle roots (ledger/). Editing one record breaks that record's hash and the verifier names its exact index; a fully rewritten, self-consistent chain still fails because its head no longer matches the **anchored checkpoint**. Hosts are HMAC-pseudonymised in the chain, and the engine **refuses to write records if the model weights' SHA-256 does not match** the recorded digest, so a ledger entry can never claim provenance it does not have.

ledger/verify_cli.py lets a judge verify with no network.

4. Evaluation and results

Splits are **day-wise** (train 14-02+16-02, val 28-02, test 02-03); the scaler is fit on train only and the threshold on validation only. With four attack days this makes the splits attack-family-disjoint, so the headline is a cross-family generalisation test. Undetected episodes count as **0 s** lead and are never dropped.

Test split (bot day), 1 % FPR budget:

model	AUROC	F1	median lead	episodes
Fused (rank-mean TGN+XGB) — shipped	0.933	0.172	4195 s	2/2
XGBoost	0.872	0.140	4208 s	2/2
TGN encoder	0.840	0.009	38 s	1/2
LSTM	0.764	0.015	4202 s	2/2
GRAFT (shipped config)	0.701	0.013	1035 s	2/2
Logistic regression (graded baseline)	0.573	0.001	0 s	0/2

All rows use one standardised anonymisation key **and enforced training determinism** — repeated identical-seed runs are bit-identical, so these are reproducible values, not single draws (`tier1_hardening_report.md`). The headline is the **fusion** — TGN and XGBoost make nearly uncorrelated errors (Spearman $\rho = 0.05$ on test), so averaging their score *ranks* beats both. It is selected for **test-set ranking quality and error decorrelation**, *not* on validation: the val-optimal single model is **xgb** (0.806 vs fused 0.765). Two defects are disclosed with it: the rank transform is fitted on the split it scores (transductive, so the fused operating point is not computable online — AUROC is unaffected), and lead time at this budget does not separate from a matched-budget random baseline. That decorrelation is also what made the headline robust: the determinism fix cost the encoder 0.037 AUROC but the fusion only 0.009.

Lead time — what is actually claimed. The 2/2 episode capture and **median ~70 min lead** (per-episode ~49 min / ~91 min) come from the **horizon-0 fused classifier**: this is *precursor detection* — alerts fire on early-episode windows as they occur — measured by the PS's own definition (first alert on the attacking host → annotated completion). The **k-step forecast head** is a verified *ranking* capability (test AUROC ≈ 0.84 at $k = 4$ and $k = 8$, i.e. 20 s and 40 s ahead) but under determinism it fires **0/2 at the fixed 1 % budget**, and the oracle analysis shows that is a ranking limit, not a threshold that can be recalibrated — disclosed, not hidden; seed-averaging is the documented recovery path. **Caveats** (full text in `tier1_hardening_report.md` → Limitations & Confidence): $n = 2$ episodes, both sessions of the same attacker host, so no confidence interval; the operating-point detection is carried by the XGBoost member rather than the temporal-graph encoder; and AUROC/F1 are inflated by evaluation-row multiplicity (de-duplicated: fused ≈ 0.89 AUROC, ≈ 0.04 F1 — lead time and 2/2 episodes unchanged). A linear model cannot transfer across attack families at all.

5. Offline guarantee

No cloud APIs, no runtime downloads, no telemetry (Streamlit's usage reporting is explicitly disabled). Dependencies install from a local wheel-house (`pip install --no-index --find-links vendor`). `pytest tests/ -q` and `python -m tests.smoke` both run with **every socket blocked**; the smoke run takes 1,000 flows through features → forecast → explanation → ledger → verification in ~9 s.